January 27, 2023

DSML : CC Maths.

Problem Solving Lecture - 2.

Interview Style !!

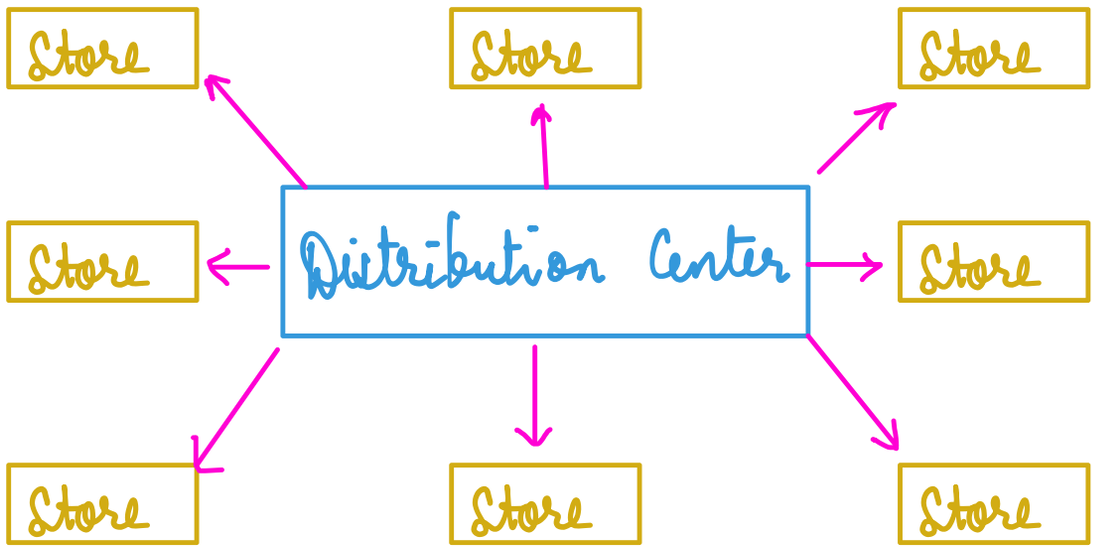
Topics :

(a) Probability - Basic Definitions.
(b) Bayes' theorem.
(c) Combinatorics.
(d) Descriptive stats.
(e) Distributions.
(f) Central limit theorem
(g) Confidence intervals.

Volunteers Requested !! → Faruk.

Class begins @ 9:05 p.m.

Q1] Supply Chain.



You have been provided with the weekly sales data for a thousand stores.

Q2] Airline Overbooking.

5% of people making a reservation don't show up. Suppose we sell 52 tickets for a flight which holds 50 passengers.

What is the probability there will be a seat for every passenger who shows up?

$n = 52$.

$p = 0.05$

$k = 50$

X : no. of people who don't show up

$P[X > 1]$

1 — 0.05

52 — ?

$\lambda = 0.05 \times 52$.

(b) 1 − binom.cdf(k = 1, n = 52, p = 0.05)

1 − poisson.cdf(mu = 0.05 × 52, k = 1)

Q 3] Pooled Blood Tests.

We test the blood of 4 people at a time for diseases. If clean, store all 4.
If diseased, conduct 4 more tests.
The probability of an individual sample being diseased is 0.1. Find the expected number of tests.

binom.pmf(k=0, n=4, p=0.1).

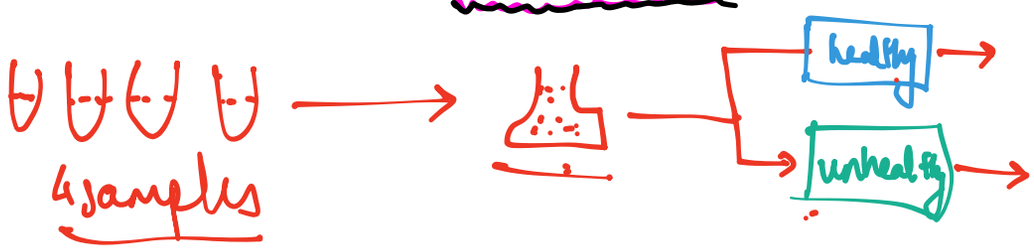
1 − binom.pmf(k=0, n=4, p=0.1)

P[diseased] = 0.1.

n = 4.

✓ X: number of tests to conduct → 1, 5

$$E[X] = 1 \cdot P[X=1] + 5 \cdot P[X=5].$$



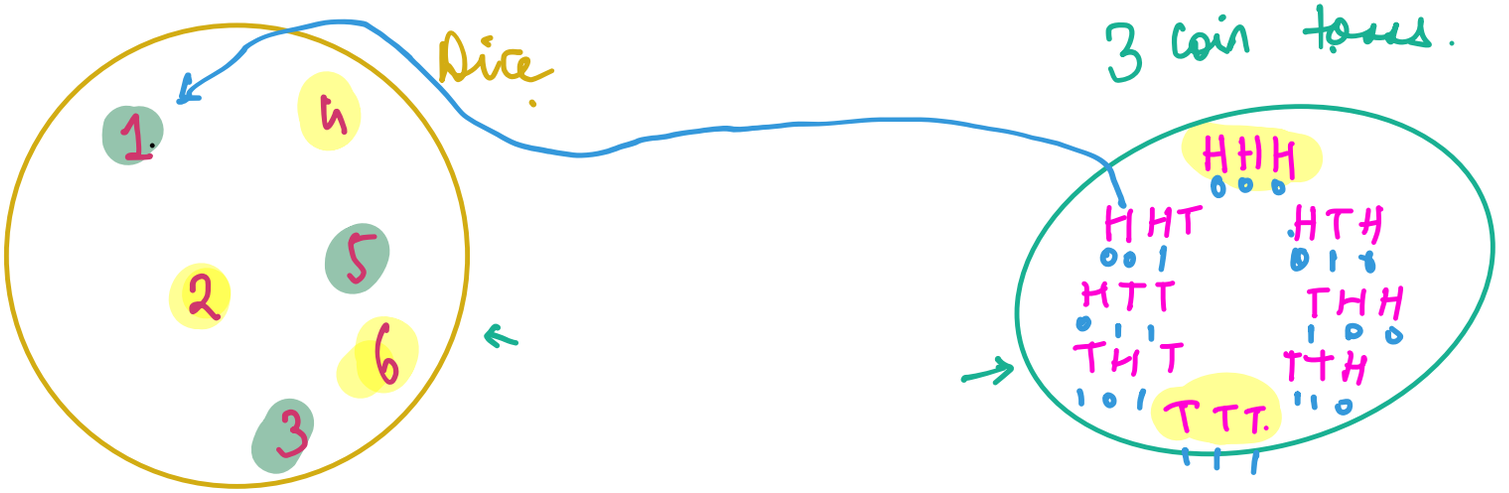
healthy → Accept all 4 → 1 test

unhealthy → Test all 4 again → 5 tests.

binom.pmf(k=0, n=4, p=0.1).

$$\downarrow$$

$$\underbrace{{}^{4}C_{0}}_{\downarrow\ 1} \times \underbrace{(0.1)^{0}}_{\downarrow\ 1} \times \underbrace{(0.9)^{4-0}}_{\underline{(0.9)^{4}}}$$



$H \rightarrow 0$

$T \rightarrow 1$

```
if "H":
    return np.random.choice([1,2,3])
else:
    return np.random.choice([4,5,6])
```

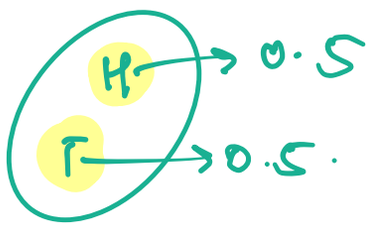
Q] Given a biased coin:

$$\begin{cases} P(H) = 0.7. \\ P(T) = 0.3. \end{cases}$$

[H, H, H, H, H, H, H, T, T, T]

Can you simulate a fair coin from a biased coin?

Biased coin



$P(H)$

$P(\{HT\} \mid \{HT, TH\})$

$P(\{TH\} \mid \{HT, TH\})$

$$\{H, T\} \xrightarrow{2} \{HH, HT, TH, TT\}$$

$H \to 0.7$, $T \to 0.3$

$HH \to 0.49$, $HT \to 0.21$, $TH \to 0.21$, $TT \to 0.09$

$$\frac{P(\{HT\} \cap \{HT, TH\})}{P(\{HT, TH\})} = \frac{P(\{HT\})}{P(\{HT, TH\})} = \frac{0.21}{0.42} = \frac{1}{2}\ ?$$

Q5] Combinatorics. (H.W.).

Out of 4 friends, who were all born in the same non-leap year:

(a) Probability that all have the same b'day?
(b) Probability that all have different b'days?
(c) Probability that 2 have the same b day?
(d) Probability that 3 have the same b day?

$$P(\text{All have the same b'day}) = \frac{(365)}{(365)^4} = \frac{1}{(365)^3}$$

Hint: sum of probabilities should be 1.

$$\underset{365}{\underset{\uparrow}{_}} \quad \underset{365}{\underset{\uparrow}{_}} \quad \underset{365}{\underset{\uparrow}{_}} \quad \underset{365}{\underset{\uparrow}{_}}$$

$P(H) = 0.7$

$P(T) = 0.3$

$\{H, T\} \longrightarrow \{HH, HT, TH, TT\}$

$H \to 0.7$, $T \to 0.3$

$HH \to 0.49$ (Repeat), $TT \to 0.09$ (Repeat)

$HT \to 0.3 \times 0.7$, $TH \to 0.3 \times 0.7$

H_1, T_1

$$P(H_1) = P(\{HT\} \mid \{HT, TH\})$$

$$= \frac{P(\{HT\} \cap \{HT, TH\})}{P(\{HT, TH\})}$$

$$= \frac{P(\{HT\})}{P(\{HT, TH\})} = \frac{\cancel{0.21}}{\cancel{0.21} \times 2} = \frac{1}{2}.$$